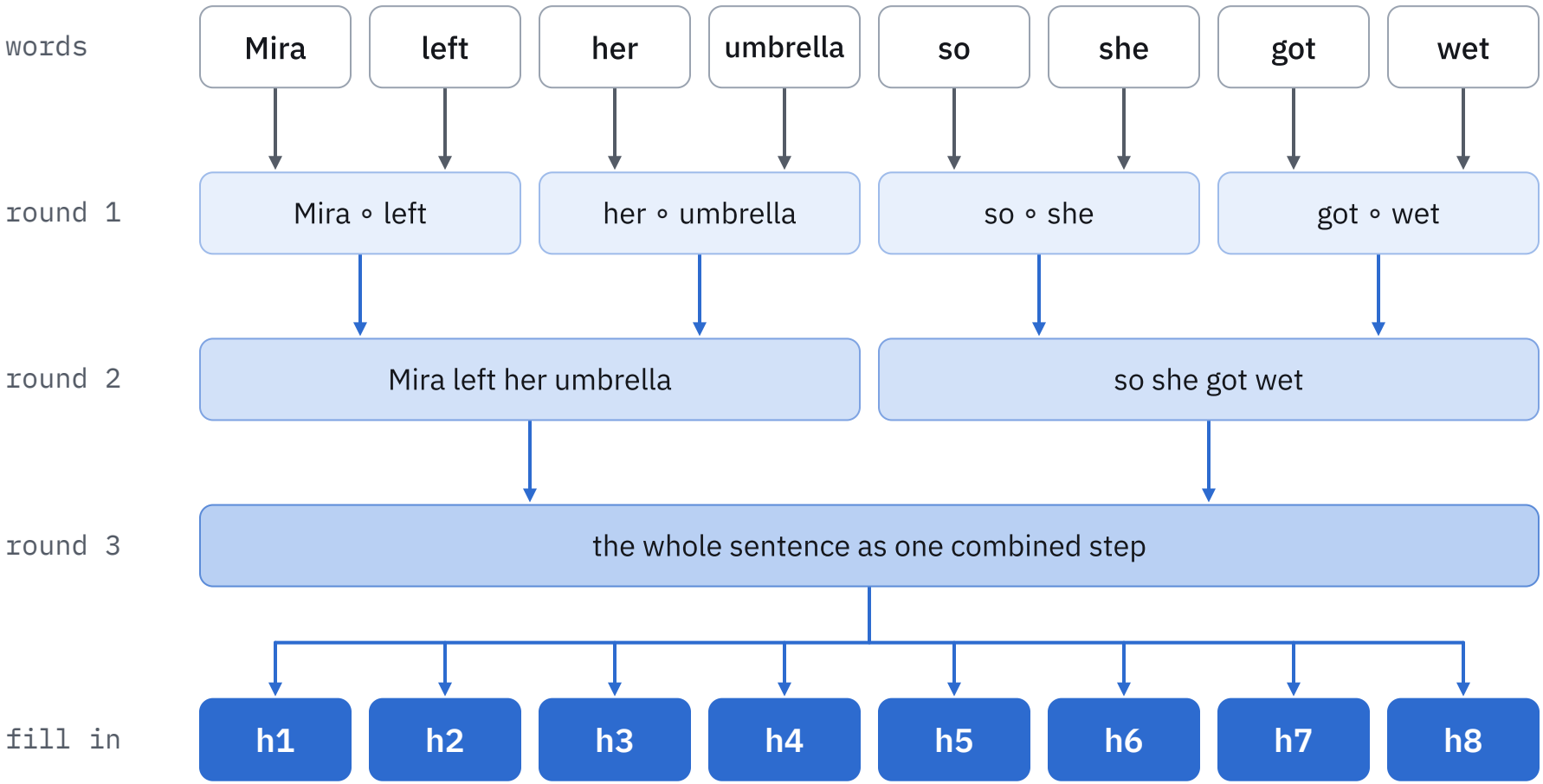
FIGURE 6

The parallel scan: eight words, three rounds



Each step is “multiply, then add”, so two neighbouring steps fuse into one step of the same form, and the grouping doesn’t change the answer. Fuse pairs, then pairs of pairs, then fill in the state after every word: 8 words take 3 rounds, 1,000 take 10.